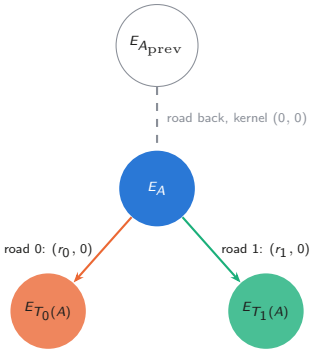
Isogenies on Oasis

Qiskit Fall Fest 2026, Track 1

Three Qiskiteers

November 2026

The campus



- A spot is a curve $E_A: \; y^2 = x^3 + Ax^2 + x$, with $A \in \mathbb{F} = \mathbb{F}_p(i)$, $p \equiv 3 \pmod 4$.
- A road is a 2-isogeny. Its kernel is one 2-torsion point $(r, 0)$:

$$r_{0,1} = \frac{-A \pm \sqrt{A^2 - 4}}{2}, \quad \text{sorted by } \operatorname{key}(u + vi) = u + pv.$$

- Bit b takes the road through r_b:

$$T_b(A) = \operatorname{can}(2 - 4r_b^2), \quad \operatorname{can}(z) = \text{the smaller key of } \pm z.$$

- A password is a route $x = b_1 \cdots b_L$. It ends at

$$H(x) = j(E_{A_L}) = \frac{256(A_L^2 - 3)^3}{A_L^2 - 4}.$$

- Small campus: $p = 1019$, $A_0 = 977 + 297i$, $L = 12$, $j^* = 797 + 921i$.

Constraints

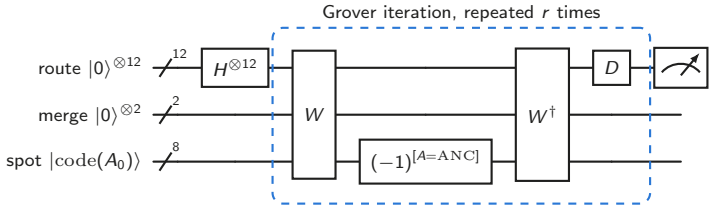
Search rules

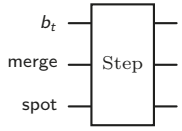
- The survey walks each road once (at most 2 moves per spot). It does not use A_0, L or j^*.
- After the survey: no path counting, no search backwards from ANC, no meet-in-the-middle, no route lists.
- The only classical work: walk one measured route once, to check it.
- The test "are we at ANC?" reads the spot inside the circuit. It is not built from H.
- If no route gets to ANC, the code says so.

Build rules

- All circuits are built and run in Qiskit.
- Resources are given after transpiling to a stated basis: queries, qubits, two-qubit gates, depth. Part III also gives the Toffoli and T counts.
- One command reproduces every result. It takes p, A_0, L, j^*.
- One query is one use of the one-step circuit.

Part I: the circuit

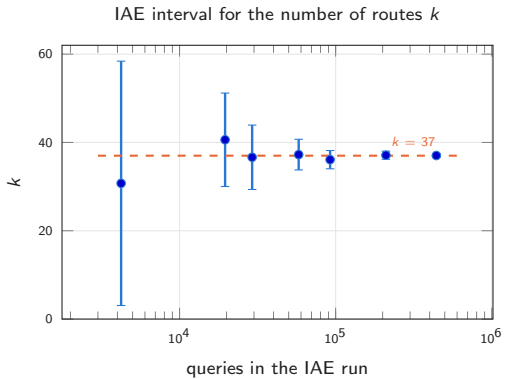




$W = \mathrm{Step}_{12} \cdots \mathrm{Step}_1$, step t reads bit b_t

- Survey: 255 spots, 510 roads. Each spot gets an 8-bit code.
- Two merge qubits make the step one-to-one, so it is a permutation. It uses 17 single-target gates (De Vos–Van Rentergem).
- Oracle $= W$, phase on the ANC codes, $W^\dagger$: 24 queries.
- 22 qubits. In $\{cx, rz, sx, x\}$: one step 13,395 CX, depth 42,173.

Part I: counting the routes



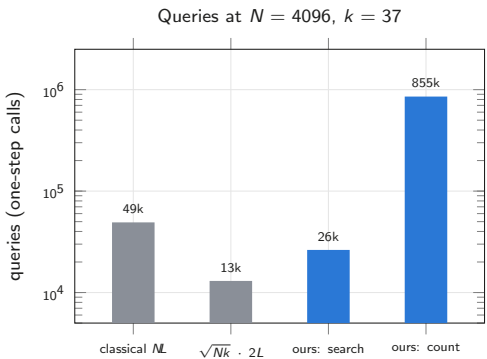
- **IAE** (Iterative Amplitude Estimation): it estimates $a = k/N$ from circuits $Q^m A|0\rangle$ with Grover powers Q^m. It needs no QFT. Each round makes the confidence interval smaller (Appendix A1).
- We stop when the interval for $k = aN$ holds one integer.
- Result: $k = 37$ at confidence ≥ 0.99.
- $k = 0$ gives an interval at 0. The code then reports "no route".

Part I: finding every route

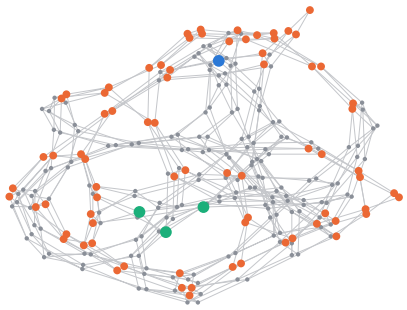
- Grover with $r = 8$ iterations per run, 137 runs. Each measured route is checked once by one classical walk (1 run rejected).
- All 37 routes are found. They end at 3 spots:

A_{12}	routes	j
$3 + 10i$	14	$797 + 921i$
$933 + 205i$	9	$797 + 921i$
$561 + 411i$	14	$797 + 921i$

- The 0.99-confidence count costs most of the queries. The scale is $\sqrt{Nk}$ for all routes.



Part II: the target



$p = 1019$: 255 spots. ● on the tower (75), ● ANC (3), ● A_0

- Top of the Clock Tower: $T_p = \{E : j(E) \in \mathbb{F}_p\}$.
- Deuring–Eichler: $|T_p| = 2h(-p)$ or $h(-p)$, where h is a class number.
- Siegel: $h(-p) = p^{1/2+o(1)}$. All spots: $|V_p| = p/12 + O(1)$.

$$\frac{|T_p|}{|V_p|} = p^{-1/2+o(1)}$$

- Small campus: 75 of 255 spots (29%). At $p \approx 2^{256}$: about 1 spot in 2^{125}.

Part II: from spots to routes

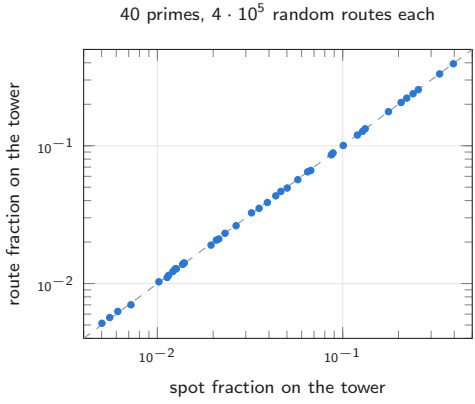
- A route is a walk without backtracking on a 3-regular Ramanujan graph (Pizer).
- Lubetzky–Peres: the end of the walk is close to uniform,

$$d_{\mathrm{TV}} \le O\big(\sqrt{p}\, 2^{-L/2} L\big).$$

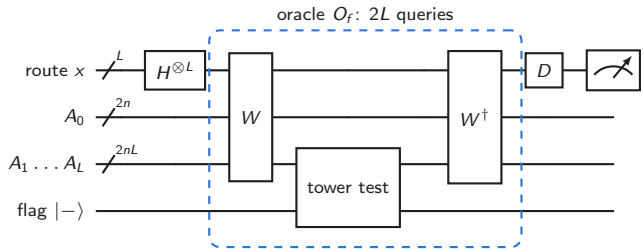
- With $L = \lceil 3 \log_2 p \rceil$ this is $\widetilde{O}(p^{-1})$, much less than the target fraction. So

$$q_L = \Pr_x[f(x) = 1] = p^{-1/2+o(1)}.$$

- Data (right): route fraction = spot fraction, for 40 primes, already with $L = 2\lceil \log_2 p \rceil$.



Part II: the upper bound



- $O_f |x\rangle = (-1)^{f(x)} |x\rangle$, $f(x) = [j(E_{A_L(x)}) \in \mathbb{F}_p]$.
- With $\sin^2 \theta = q_L$: after t iterations, $P(t) = \sin^2((2t+1)\theta)$.
- $t \approx \frac{\pi}{4\sqrt{q_L}} = p^{1/4+o(1)}$ iterations, each $2L = O(\log p)$ queries.
- q_L unknown: BBHT, at most $\frac{9}{4}\sqrt{1/q_L}$ on average.
- Classical random routes: $1/q_L = p^{1/2+o(1)}$.

Part II: the lower bound

Black-box model

Input: an unknown $f : \{0,1\}^L \to \{0,1\}$ with k marked inputs. The only access is the oracle $O_f |x, z\rangle = |x, z \oplus f(x)\rangle$.

BBBV, Zalka: every quantum algorithm with constant success needs

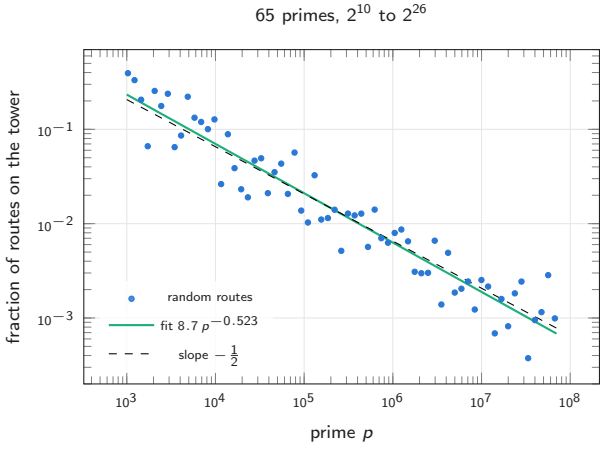
$$\Omega\left(\sqrt{N/k}\right) = \Omega\left(q_L^{-1/2}\right) = \Omega\left(p^{1/4-o(1)}\right)$$

oracle queries. Grover meets this bound.

The real arithmetic

- The step is explicit field arithmetic. An algorithm can use this structure, so the black-box bound does not apply.
- "A route to the tower exists" is in NP (Appendix A3). A proof that no polynomial-time quantum algorithm exists would show NP $\not\subseteq$ BQP, and so P $\neq$ NP.
- No known algorithm beats $p^{1/4}$ under these rules (Appendix A4).

Part II: data



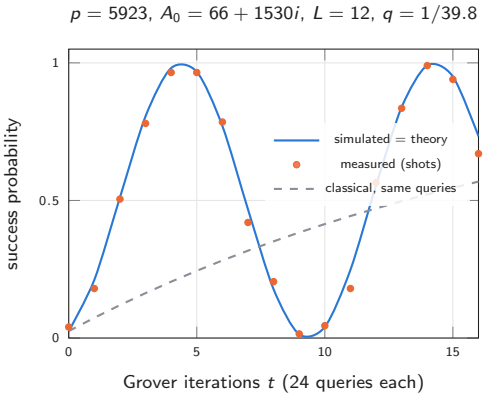
- Fit: $\widehat{q} \approx 8.7\,p^{-\alpha}$,

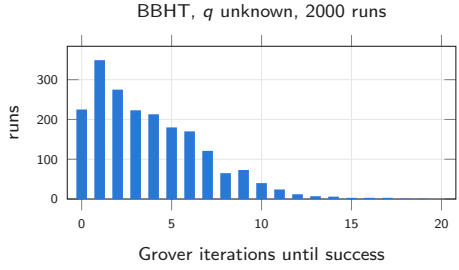
$$\widehat{\alpha} = 0.523 \pm 0.040$$

(65 primes, 2 standard errors).

- Exact spot fractions, 40 primes: $\widehat{\alpha} = 0.560 \pm 0.081$.
- The scatter comes from the class number $h(-p)$, not from noise: the same primes are high in both columns.
- Theory: $\alpha = \frac{1}{2}$.

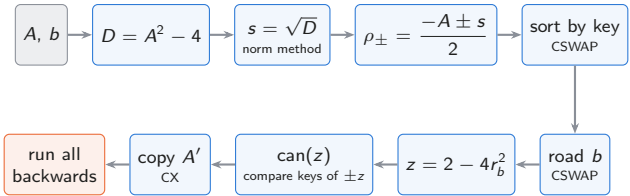
Part II: demonstration





- Simulation = theory to 4 decimals. At $t = 4$: 0.981, against 0.205 classically.
- BBHT: on average 91 queries and 7.9 checks. Classical: 477 steps.
- 25 qubits. One iteration in $\{cx, rz, sx, x\}$: $3.40 \cdot 10^6$ CX, depth $1.11 \cdot 10^7$.

Part III: one step from arithmetic



Square root from the norm

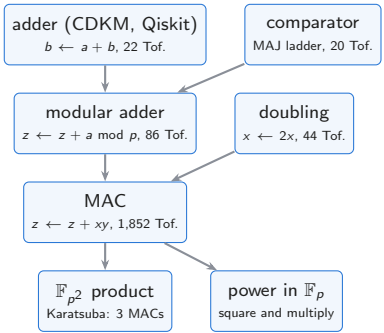
$D = \alpha + \beta i$. Two exponentiations in $\mathbb{F}_p$:

$$\lambda = (\alpha^2 + \beta^2)^{\frac{p+1}{4}}, \quad \delta = \tfrac{\alpha+\lambda}{2}, \quad r = \delta^{\frac{p-3}{4}}.$$

$y = r\delta$, $t = \beta r/2$, $q = [yr = 1]$: $s = y + ti$ if $q = 1$, else $s = -t + yi$. No inversion (Appendix A5).

- No branches: every "if" is a flag qubit and controlled gates.
- Any root s works: the key sort fixes the order of $\rho_\pm$.
- $|A, b\rangle|0\rangle \mapsto |A, b\rangle|T_b(A)\rangle$, all work qubits back to 0 (Bennett).
- Clock-Tower test without division: $j \in \mathbb{F}_p \iff \overline{N}D = N\overline{D}$.

Part III: building blocks



Toffoli counts for $p = 1019$, $n = 10$ bits

- Gates: X, CX, Toffoli, CSWAP only.
- Modular adder: add, subtract p, add p back if the sign bit is 1, then clear the flag with the comparator.
- Doubling is a relabelling of qubits plus one subtraction of p.
- MAC: for each bit y_i, add $y_i \cdot x$, then double x. This gives n modular adders, so $O(n^2)$.
- An exponentiation is $O(n)$ products, so one step is $O(n^3)$.

Part III: checks at $p = 1019$

Every input

- Aer matrix-product-state simulator, 587 qubits.
- Step: 510 inputs (255 spots, both bits). Output = classical $T_b(A)$, inputs unchanged, every work qubit 0. **510/510.**
- Clock-Tower test: **255/255** spots.
- 32 inputs on random spots at $p = 46351$: all correct.

Why basis states are enough

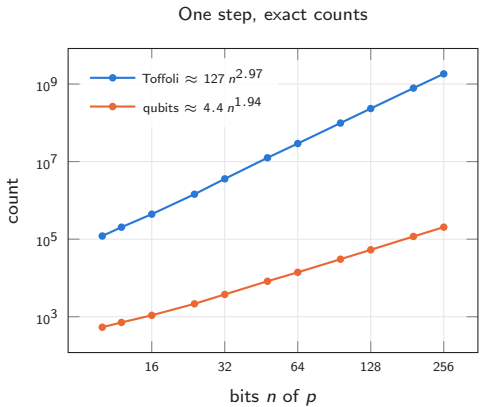
X, CX, Toffoli and CSWAP send basis states to basis states, so the circuit U is a permutation matrix. If $U|x\rangle|0\rangle = |x\rangle|f(x)\rangle$ for every x, then by linearity

$$U \sum_x c_x |x\rangle|0\rangle = \sum_x c_x |x\rangle|f(x)\rangle .$$

Phases

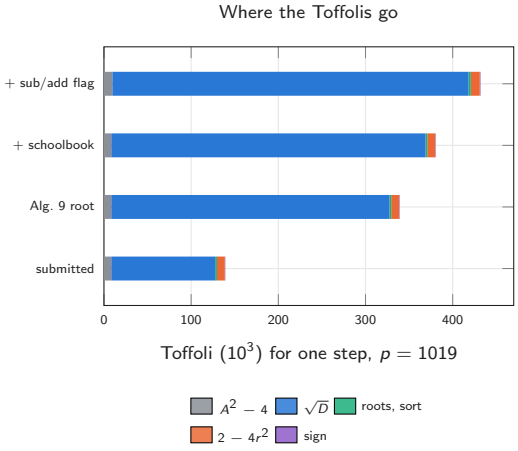
64 states $\cos w\,|x\rangle + e^{i\phi} \sin w\,|y\rangle$ go through the circuit and then the inverse of a reference circuit. All return to $|0\rangle$. A deliberate phase error is detected.

Part III: cost at $p \approx 2^{256}$



one step, $n = 256$	
Toffoli	$1.83 \cdot 10^9$
T (7 per Toffoli)	$1.28 \cdot 10^{10}$
qubits	204,049
Part II search, $L = 2n = 512$	
target fraction q	$7/\sqrt{p} = 2^{-124.9}$
Grover iterations	$2^{62.1}$
Toffoli	$2^{102.9}$
two-qubit, $\{cx, rz, sx, x\}$	$2^{105.9}$
depth	$2^{104.4}$
logical qubits	466,000
classical random routes	$2^{133.9}$ steps

Part III: trade-offs, other algorithms



Each row changes one algorithm of the submitted circuit:

change	Toffoli	factor
submitted	139,236	1.00
root: Algorithm 9 ($\mathbb{F}_{p^2}$ powers)	339,162	2.44
+ schoolbook $\mathbb{F}_{p^2}$ product	380,506	2.73
+ flag by subtract and add back	431,914	3.10

- The root is 86% of the step. In $\mathbb{F}_p$ a product is 1 MAC; in $\mathbb{F}_{p^2}$ it is 3 or 4.
- A comparison needs only the carry: $2n$ Toffolis, not $4(n+1)$.
- At $n = 256$: submitted $= 0.34\times$ the slowest version.

Part III: trade-offs, qubits and depth

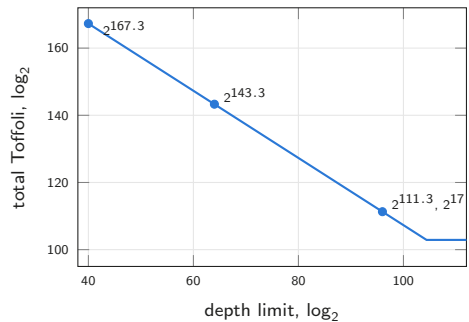
Qubits: pebbling

Bennett keeps each intermediate value: about $1.5n$ registers of n qubits per exponentiation, so qubits $\sim n^2$.

strategy	registers	steps run
Bennett (ours)	T	$2T$
$\sqrt{T}$ checkpoints	$2\sqrt{T}$	$\approx 3T$
recursive halving	$\log_2 T$	$T^{1.58}$

We keep Bennett: it gives the fewest gates.

Part II search at $p \approx 2^{256}$ with a depth limit



With depth limit 2^m: $S = (2^{104.4-m})^2$ machines; total $\times\sqrt{S}$.

Part IV: the Djinns

What still works

- The graph is the same: degree, roads, cycles, expansion.
- Classical: random walks ($p^{1/2}$ tower tests), BFS/DFS with a table of names, cycle and collision search.
- Our Part I method: the survey asks the Djinn for every road ($O(N)$), then the step circuit is built from the survey.
- With a *coherent* Djinn and volunteer: Grover and quantum walks still give $p^{1/4}$.

What stops working

- Arithmetic on A: the Part III step circuit, the j test, Frobenius.
- Methods from number theory (class groups, endomorphism rings).
- A human Djinn or volunteer answers one question at a time. No superposition, so no Grover.

Lower bound

The names are random, so the volunteer is close to a black box. The $\Omega(p^{1/4})$ bound needs one more condition: the tower set must not be visible from the graph shape.

Summary

- **Part I:** all 37 passwords, k counted at confidence ≥ 0.99, survey rules kept.
- **Part II:** $p^{1/4+o(1)}$ quantum against $p^{1/2+o(1)}$ classical. Black-box lower bound; a lower bound for the real arithmetic would prove P $\neq$ NP. Fitted exponent 0.523 ± 0.040; Grover demo $=$ theory.
- **Part III:** one step from arithmetic only, 510/510 correct. At 2^{256}: $1.8 \cdot 10^9$ Toffolis per step, 2^{103} for the search.
- **Part IV:** the speed-up needs coherent access, not the names.

Reproduce

```
uv run oasis --part 1 --p 1019 \
  --A0 "977 + 297i" --L 12 \
  --jstar "797 + 921i"
uv run oasis --part 2
uv run oasis --part 3
```

Results: `results/`

A1. Iterative Amplitude Estimation

- Problem: $A|0\rangle = \sqrt{a}\,|\text{good}\rangle + \sqrt{1-a}\,|\text{bad}\rangle$. Estimate $a = \sin^2\theta_a$.
- Grover operator $Q = AS_0A^\dagger S_{\text{good}}$. Then $Q^mA|0\rangle$ is good with probability $\sin^2((2m+1)\theta_a)$.
- Each round picks the largest power m that keeps $(2m+1)\theta_a$ inside one half-period of the current interval. It measures, and narrows the interval for θ_a (Clopper–Pearson).
- Cost: $O\big(\frac{1}{\varepsilon}\log\frac{2}{\alpha}\log\frac{\pi}{4\varepsilon}\big)$ queries for error ε at confidence $1-\alpha$, with no QFT (Grinko, Gacon, Zoufal, Woerner 2021).
- Here: good = spot register at an ANC code, $a = k/N$. We halve ε until the interval for $k = aN$ holds one integer; $\alpha = 0.01$.

A2. Theorems we use

Deuring–Eichler For $p \equiv 3 \pmod 4$, the number of supersingular $j \in \mathbb{F}_p$ is $2h(-p)$ if $p \equiv 3 \pmod 8$ and $h(-p)$ if $p \equiv 7 \pmod 8$.

Siegel $h(-p) = p^{1/2+o(1)}$.

Pizer The supersingular 2-isogeny graph is 3-regular and Ramanujan: every nontrivial eigenvalue has $|\lambda| \le 2\sqrt{2}$.

Lubetzky–Peres For a walk without backtracking of length L on a 3-regular Ramanujan graph with N_E directed edges: $\left\|\tilde{\mu}_L/\pi_E - 1\right\|_{L^2}^2 \le 2N_E 2^{-L}(8L^2+1)$.

BBBV, Zalka Search for one of k marked items among N with a black-box oracle needs $\Omega(\sqrt{N/k})$ queries; Grover is optimal.

BBHT With k unknown and $\lambda = 6/5$, the expected number of Grover iterations is at most $\frac{9}{4}\sqrt{N/k}$.

A3. The Clock-Tower problem is in NP

Decision problem: given p and A_0, is there a route of length $L = O(\log p)$ that ends on the Clock Tower?

1. **Certificate:** the route $x = b_1 \cdots b_L$. Its length is $O(\log p)$ bits.
2. **Check:** L steps, each a fixed number of operations in $\mathbb{F}_{p^2}$ (one square root is one exponentiation), then compute j and test $\mathrm{Im}\, j = 0$. Each operation costs $\mathrm{poly}(\log p)$, so the check costs $\mathrm{poly}(\log p)$.
3. So the problem is in NP. The input size is $n = \log_2 p$.
4. Our cost $p^{1/4} = 2^{n/4}$ is exponential in n. A proof that no polynomial-time quantum algorithm exists would show NP $\not\subseteq$ BQP. Since P $\subseteq$ BQP, this gives P $\neq$ NP.
5. So an unconditional lower bound for the real arithmetic is out of reach today. Lower bounds exist only in a model with a black-box oracle.

A4. Known algorithms that use the arithmetic

algorithm	solves	cost	why not here
Biasse–Jao–Sankar (2014)	path between two curves, quantum	$\widetilde{O}(p^{1/4})$	same exponent as Grover
Page–Wesolowski (2023)	path, classical, from both ends	$\widetilde{O}(p^{1/2})$	meet-in-the-middle; forbidden by the rules
Delgado (2026)	one endomorphism, classical	$p^{1/3+o(1)}$	not a fixed-degree path; the path version needs GRH and gives longer paths
Wesolowski (2021)	path $\Leftrightarrow$ endomorphism ring	reduction	assumes GRH

None of these gives less than $p^{1/4}$ for a route of length $\Theta(\log p)$ from A_0 under the rules.

A5. Proof of the square root from the norm

Let $s = x_0 + x_1 i$ with $s^2 = D = \alpha + \beta i$. Then $x_0^2 - x_1^2 = \alpha$, $2x_0x_1 = \beta$, and $(x_0^2 + x_1^2)^2 = \alpha^2 + \beta^2 = m$.

- m is a square in $\mathbb{F}_p$, so $\lambda = m^{(p+1)/4}$ gives $\lambda^2 = m$, and $x_0^2 + x_1^2 = \pm\lambda$.
- $\delta = (\alpha + \lambda)/2$ and $\delta' = (\alpha - \lambda)/2$ satisfy $\delta\delta' = -\beta^2/4$. Since -1 is not a square ($p \equiv 3 \bmod 4$), exactly one of them is a square (if $\beta \neq 0$).
- $r = \delta^{(p-3)/4}$, $y = r\delta$: then $y^2 = \delta^{(p+1)/2} = \pm\delta$ and $yr = \delta^{(p-1)/2} = \pm 1 = q$-flag.
- $q = 1$: $y^2 = \delta = x_0^2$, $x_1 = \beta/(2y) = \beta r/2 = t$, so $s = y + ti$.
- $q = 0$: $y^2 = -\delta = x_1^2$, $x_0 = \beta/(2y) = -\beta r/2 = -t$, so $s = -t + yi$.
- $\delta = 0$ only if $\beta = 0$ and α is not a square; then $\delta := \alpha$ gives $q = 0$, $s = yi$ with $y^2 = -\alpha$.

Checked on every nonzero square for 13 primes ≤ 83, and in the circuit on random squares at $p = 43$ and $p = 131$.

A6. Resources

Part I, $p = 1019$, $L = 12$, $\{cx, rz, sx, x\}$, level 1

	queries	CX	depth
one step	1	13,395	42,173
Grover iteration	24	319,106	1,009,282

22 qubits. Total: 854,784 queries to count, 26,304 to search.

Part II, $p = 5923$, $L = 12$, $\{cx, rz, sx, x\}$, level 1

	queries	CX	depth
one step	1	141,813	462,568
Grover iteration	24	3,401,466	11,100,684
search, $t = 4$	96	13,605,864	44,402,739

25 qubits.

Part III, $p = 1019$, $\{x, cx, ccx\}$; two-qubit in $\{cx, rz, sx, x\}$ = CX +6·Toffoli

	qubits	Toffoli	2-qubit	depth
step	587	139,236	1,127,164	307,933
tower test	192	37,794	–	63,148

T: 974,652 (7 per Toffoli) or 556,944 (4 per Toffoli).

Part III, $n = 256$

step: Toffoli / qubits	$1.83 \cdot 10^9$ / 204,049
tower test: Toffoli	$2.1 \cdot 10^7$
search: Toffoli / depth	$2^{102.9}$ / $2^{104.4}$

A7. Note on the phase test

- The test prepares $\cos w\,|0\rangle\,|x\rangle + e^{i\phi}\sin w\,|1\rangle\,|y\rangle$, runs the new circuit, then the inverse of the reference circuit, then undoes the preparation. The result is $|0\rangle$ with probability $1 - \sin^2(2w)\sin^2(\Delta\theta/2)$.
- A path of pairs through the inputs gives $\theta(x) = \theta(y)$ for all of them: the two circuits are equal up to one global phase.
- With equal amplitudes ($w = \pi/4$), Aer's matrix-product-state simulator lost the relative phase for some values of ϕ, also for a circuit followed by its own inverse. Two equal Schmidt values make the SVD degenerate.
- With $w = 0.55$ every case passes. A deliberate error (Z on b) is found on every pair with different b, and only there.